

ngspice-46 vs. Spectre — Feature Gap Analysis

Ngspice + OpenVAF Enhancements project

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A capability comparison of the ngspice-46 build in this repository against a Spectre-class commercial simulator, to show where the open-source tool already matches the commercial one and where the gaps are. Grounded in a direct read of the ngspice-46/src/ tree (analyses, solver, devices, RF, convergence, parallelism), not marketing feature lists.

Legend: ✓ present / on par · △partial, experimental, or via a workaround · ✗ absent. "Spectre" stands in for the commercial reference (Spectre / SpectreRF / Spectre X / RelXpert feature set).

The one column that matters is ngspice — Spectre has essentially everything, so its column is a baseline of ✓. The point of the table is where ngspice's mark is △or ✗.

Context: the Verilog-A / OSDI device side is not a gap — thanks to the OpenVAF-reloaded compiler in this repository it is on par with (often ahead of) commercial Verilog-A support. The gaps below are all on the simulator/analysis side.

Core numerics

Category	Feature	ngspice	Spectre
Core solver	Sparse LU (KLU) direct solver	✓ ¹	✓
Core solver	Legacy Sparse1.3 fallback	✓	✓
Core solver	Parallel / partitioned / GPU linear solve	✗	✓
Core solver	Matrix reordering + scaling beyond KLU defaults	✓	✓
Integration	Trapezoidal + variable-order Gear	✓	✓
Integration	Advanced LTE-based step/order control	✓	✓
Convergence	gmin stepping + source stepping homotopy	✓	✓
Convergence	Pseudo-transient / dynamic-gmin continuation	✓	✓
Convergence	Damped / trust-region (globalized) Newton	✓	✓
Convergence	Coordinated accuracy presets (errpreset)	✓	✓

The "matrix reordering + scaling beyond KLU defaults" row is ✓ since [Enhancement-152](#): KLU's fill-reducing ordering (`klu_ordering=amd|colamd`), row scaling (`klu_scale=none|sum|max`), and BTF permutation (`klu_btf=on|off`) are now `.options` (previously hard-coded to AMD/max/on), and the broken `klu_memgrow_factor` is fixed. They change only how the matrix factors, not the solution.

The integration methods themselves are now certified exactly ([Enhancement-181](#)): at every Gear order 1–6 the accepted trajectory satisfies the exact variable-step BDF-k formula to machine precision ($\approx 1.3e-13$) — the first direct verification of the NIcomCof coefficients, including the orders 3–6 that were dead code for ~30 years before E-128. The audit ships `.options ordfix=K` (a fixed-order verification mode) and referees the rest of this table: solver precision tracks conditioning theory under both solvers, all DC convergence-aid paths agree on a hard-DC chain, and the `xmu/lvltim=1` legacy paths work. Three apparent bugs found on the way were real numerics — the LTE order preference inverts at loose

tolerance, order-1 starters impose an $O(h^2)$ floor, and lossless-LC divergence at high order is the textbook $A(\alpha)$ stability wedge.

The Integration “advanced LTE-based step/order control” row was [△](#)because, although ngspice implements Gear coefficients for orders 1–6 (NIcomCof) and an LTE-limited-step estimate at any order (CKTtrunc/CKTterr), the stock transient controller in `dctran.c` only ever toggles the order between 1 and 2 — orders 3–6 were dead code on every ordinary run. [Enhancement-128](#) closes it with `.option dynorder`: each step it compares the raw (uncapped) LTE-limited step at the current order and its ± 1 neighbours and moves — with hysteresis, a settling hold, and an order-dependent growth cap — toward the order the local error rewards, so the higher orders are actually used. Off by default, bounded by `maxord`, verified under both linear solvers: $3\text{--}5\times$ fewer timesteps at matched accuracy on a smooth RC decay and $8.9\times$ (and more accurate) on a smooth RLC ringdown, while a nonlinear switching circuit is left result-neutral to 5 significant figures.

These two convergence rows: ngspice’s DC path is not naked — it has per-device junction limiting (30 device families), an optional `nodedamping` step clamp, and a multi-stage homotopy cascade (`dynamic_gmin` $\rightarrow$ `new_gmin` $\rightarrow$ `spice3_gmin` $\rightarrow$ source stepping). What it lacked vs. commercial tools was a principled globalized Newton (residual-based trust-region / Armijo) and a classic pseudo-transient ($\dot{X}$ -embedded) continuation — both now added. [Enhancement-111](#) adds the former: `.option linesearch`, an Armijo backtracking line search on a new KCL-residual merit $\|F\|=\|G\cdot x-b\|$ (result-neutral, off by default) — see the [implementation write-up](#). [Enhancement-153](#) completes this row with the trust-region counterpart: `.option trustregion`, a Levenberg-Marquardt Newton that damps the Jacobian ($x_{k+1}=x_k-(J+\mu I)^{-1}F$, $\mu=\lambda\cdot\|\text{diag}(J)\|$, Marquardt-scaled) rather than just the step length, re-aiming the step to regularize an ill-conditioned Jacobian — result-neutral, off by default, verified bit-identical to plain Newton. Its honest scope: because ngspice globalizes at the device level (junction limiting) before the residual is formed, a solver-level trust-region measures zero step-rejections on typical circuits and stays inert; it is a correct regularization for the residual-overshoot cases the device-level machinery does not catch. [Enhancement-112](#) then extended it to the KLU solver — it originally ran only under Sparse 1.3 and segfaulted under `.option klu` — so the line search now works under both linear solvers, with a residual-merit sequence numerically identical between them. [Enhancement-127](#) adds the latter: `.option ptcont`, a pseudo-transient continuation that embeds $f(x)=0$ in a backward-Euler pseudo-transient $f(x)+Gps\cdot(x-x_{\text{prev}})=0$ and marches the pseudo-timestep from small to large ($Gps\rightarrow 0$). The $Gps\cdot x_{\text{prev}}$ RHS coupling makes each step follow a stable trajectory (vs a static `gmin` step), so it converges — and to the physically correct root — on stiff circuits where plain Newton overshoots; result-neutral and off by default, verified under both linear solvers.

¹ KLU is compiled in (SuiteSparse, statically linked) but is not the default in this build — the default direct solver is Sparse 1.3; KLU is selected with `.option klu`. The two agree on DC / AC / transient, and — since [Enhancement-113](#) — on noise and single-ended pole-zero as well (the KLU adjoint solve was doing a non-transposed solve, silently wrong on asymmetric matrices; now fixed), and — since [Enhancement-114](#) — on DC/AC sensitivity (`.sens`), and — since [Enhancement-115](#) — on distortion (`.disto`) too. The KLU pole-zero path was then hardened end-to-end: [Enhancement-171](#) fixed the KLU complex-plane determinant (mixed real/complex pivot products and permutation parity — silent garbage for complex roots) and [Enhancement-172](#) closed the last Sparse-only analysis: balanced/differential-output `.pz` now runs under KLU (union-pattern reservation + a merge-walk `SMPcAddCol` branch), with a full-partial-pivoting fallback that also cured far-field spurious roots — so no analysis is Sparse-only under KLU any more, and since [Enhancement-180](#) no feature is either (transient checkpoint/restart, the last holdout, was a solver-mode ordering bug in `loadstate`, not a KLU limitation). [Enhancement-173](#) added a modern alternative to the fragile Muller PZ driver altogether: `.options pzeig`, a shift-invert pencil + self-contained Francis-QR eigensolver (no LAPACK), default off. KLU remains less robust on stiff transient edges. Full behavior, defaults, and a solver-by-solver sweep of the example suite: [KLU vs. Sparse 1.3 solver notes](#).

Standard analyses (analog)

Category	Feature	ngspice	Spectre
Analyses	DC operating point / DC sweep	✓	✓
Analyses	Transient (<code>.tran</code>)	✓	✓
Analyses	AC small-signal (<code>.ac</code>)	✓	✓
Analyses	Noise, small-signal (<code>.noise</code> , <code>onoise/inoise</code> + integrated)	✓	✓
Analyses	Pole-zero (<code>.pz</code>)	✓	✓
Analyses	Transfer function (<code>.tf</code>)	✓	✓
Analyses	DC sensitivity (<code>.sens</code>)	✓	✓
Analyses	Distortion (<code>.disto</code>)	✓	✓
Analyses	Measurement / post-processing (<code>.meas</code>)	✓	✓

RF / periodic steady-state suite

Tutorial: [The ngspice RF / periodic steady-state suite \(PDF\)](#) is a beginner-friendly, worked walkthrough of every analysis below — with schematics, runnable built-in and OSDI netlists, real result plots, and the physics behind each.

Category	Feature	ngspice	Spectre
RF	S-parameter analysis + noise figure (.sp)	✓	✓
RF	Touchstone (.sNp) import / export	✓	✓
RF	N-port Touchstone device (.sNp: DC+AC+transient)	✓	✓
RF	Loop-gain stability analysis (stb: phase / gain margin)	✓	✓
RF	Periodic steady state (PSS)	✓ ¹	✓
RF	Harmonic Balance (HB)	✓ ⁷	✓
RF	Periodic / phase noise (Pnoise)	✓ ³	✓
RF	Periodic AC (PAC, conversion gain)	✓ ²	✓
RF	Periodic transfer function (PXF)	✓ ⁴	✓
RF	Periodic S-parameters (PSP)	✓ ⁵	✓
RF	Quasi-periodic / multi-tone (QPSS / QPAC)	✓ ⁶	✓
RF	Envelope following	✓ ⁸	✓

¹ PSS was  (experimental `--enable-pss` flag, so .pss was unimplemented in shipped builds, and ~230 lines of shooting-loop trace per run). Since [Enhancement-117](#) it is built by default, quiet (trace behind `set ngdebug`), and verified against the analytic AC response; [Enhancement-118](#) then made it run under both linear solvers (KLU had hung on a timestep explosion from reused refactor pivots — now a full re-factor is forced each PSS step under KLU). [Enhancement-176](#) then split the shooting into a driven mode: on circuits with time-dependent sources the oscillator-style frequency hunt (and its `steady_coeff` breakpoint flood — ~0.2 ps steps, 9.6 M timepoints on a pumped varactor, never converging) is replaced by the exact source period and a `T/psspoints` step clamp, converging in a few hundred timepoints (~1000× faster) — which is what makes the whole periodic small-signal suite below cheap enough to regression-test on every sweep. It remains a shooting method (the frequency hunt still runs for autonomous oscillators) and remains the foundation for the periodic small-signal analyses below. HB is now implemented -- see note 7 (it had shipped only as a `WITH_HB` stub).

² PAC is  (complete periodic-AC analysis). The full chain is built and verified: [Enhancement-119](#) retains the periodic operating point, [Enhancement-120](#) extracts the periodic Jacobian harmonics G_k , C_k , [Enhancement-121](#) assembles and solves the $(2M+1)N$ harmonic conversion matrix $H_{\{nm\}} = G_{\{n-m\}} + j\omega_n \cdot C_{\{n-m\}}$ (row frequency: the [E-175](#) RF audit proved the column-frequency form silently drops the parametric term $\dot{C} \cdot \delta v$ on pumped capacitances and fixed every small-signal builder — HB's residual keeps ω_m by the chain rule, now an explicit mode flag), [Enhancement-122](#) wraps it in a user-facing `.pac` command (runs PSS, sweeps the input frequency, writes a complex plot), and [Enhancement-123](#) finishes it with a netlist-referenced small-signal source stimulus (a true transfer / conversion gain) and multi-sideband output — every conversion sideband $f_{in} + k \cdot f_0$ as its own named vector (`<node>_usb<k>/<node>_lsb<k>`). Verified on the RC: unit-current sideband-0 reproduces the AC driving-point impedance and, with `V1 AC 1`, sideband-0 reproduces the low-pass transfer $1/\sqrt{1+(2\pi fRC)^2}$ across the band while the conversion sidebands sit at floating-point zero (a linear circuit does not mix). The one scale caveat: the conversion matrix is solved densely (capped for modest circuits) on the brute-force shooting PSS. The same solve is the substrate for pnoise and PXF.

³ Pnoise is  (working, stationary-source first cut). [Enhancement-124](#) adds a `.pnoise` command that folds every device's noise through the adjoint of the conversion matrix ($H^T \Psi = e_{\{out,0\}}$): it loads the sideband- k transfer into `CKTrhs/CKTirhs` and calls the existing device noise routines (`NevalSrc`, `OSDI load_noise`) once per sideband, accumulating $\Sigma_k S \cdot |\Delta\Psi_k|^2$ — so it reuses every device noise model and needs no per-device code. Verified on the RC: because the circuit is linear the conversion matrix is block-diagonal, only sideband 0 contributes, and pnoise reduces exactly to `.noise (4kTR/(1+(2\pi fRC)^2))`, matching the `.noise` reference to every printed digit). [Enhancement-126](#) then adds a cyclostationary mode (`.pnoise ... cyclo`): it evaluates each device's noise PSD at every PSS sample's bias and folds it through the time-domain adjoint transfer, averaging over the period, so a pumped device's bias-dependent noise (a diode's $2qI(t)$, a resistor's flicker $\propto |I(t)|^2$) is captured. It reduces exactly to the stationary case (and hence `.noise`) for a bias-independent source by Parseval. [Enhancement-177](#) built an independent noise-folding referee (from-scratch Python conversion matrix + TRNOISE transient Monte-Carlo + LTI/white limits): the white path is measured-correct to 6 digits, and the referee caught a real bug — the stationary pnoise/qnoise/phasenoise loops evaluated every folded sideband at the output frequency, whereas sideband- k noise originates at $|f+k \cdot f_0|$ (frequency-dependent PSDs — flicker, `noise_table` — folded through conversion were wrong; 21% high on the referee circuit, unbounded as $f \rightarrow f_0$; digit-exact proof both ways). [Enhancement-178](#) then made the cyclo mode exact for colored sources too (it had assumed a frequency-flat PSD): per-generator envelope harmonics B_q are recovered by load polarization against the sideband-

0 adjoint (five DEVnoise sweeps per orbit sample, no device-API change) and folded from their source frequencies, with the spectral shape measured pointwise. The physics: flicker sees $\frac{1}{f}m^2$, white sees $\frac{1}{f}m^2$ — only the envelope's DC feeds the $1/f$ band, so the old flat identity was 23% high ($\pi^2/8$) on $|\sin|$ -modulated flicker ($\text{onoise} \cdot f = R I^2 \cdot K F \cdot (|I|)^2$, not (I^2)) and 34% high on the referee's conversion circuit; the exact path matches the referee to $\frac{1}{3}e-4$ and reduces to the (now exact) stationary sum in the constant-envelope limit. [Enhancement-140](#) closes the gap (✓) with the oscillator phase-noise piece. *hbosc* is an autonomous harmonic balance: an oscillator has no source, so the HB residual $F(V) = I_R + [dq/dt] = 0$ is solved for the harmonics and the unknown oscillation frequency w_0 , the singular conversion matrix (its phase mode $u_k = jk \quad V_k$) bordered with dF/dw_0 and a phase gauge and Newton-solved from a transient seed. *phasenoise* then reports $L(df)$: the adjoint of the conversion matrix at OFFSET df , unit at the carrier sideband, folds the device noise to the output; as $df \rightarrow 0$ the limit-cycle matrix goes singular through the phase mode, so the folded noise diverges as $1/df^2$ — the phase-noise skirt — normalized to the carrier power. Verified 8/8 on an LC oscillator: autonomous HB converges to f_0 and the describing-function amplitude, $L(df)$ has the -20 dB/dec skirt near the carrier flattening into the noise floor, at a physical absolute level, and the thermal noise scales as $L \propto T$ (doubling T raises L by exactly 3 dB).

*4 PXF is ✓ (complete). [Enhancement-125](#) adds a `.pxf` command — the adjoint of PAC. It solves $H^T \Psi = e_{\text{out}}$, 0 per frequency and dots each sideband block of Ψ with the netlist AC-source pattern B_0 to get the input $\rightarrow$ output transfer at each sideband ($x_f, x_{f_usb<k>}, x_{f_lsb<k>}$). By the identity $(H^{-1}B)_{\text{out}} = (H^{-T}e_{\text{out}})^T B$ the sideband-0 transfer is bit-identical to the PAC response at the output — a reciprocity cross-check that pins the adjoint solve — verified on the RC to equal the analytic low-pass transfer, with the conversion sidebands at floating-point zero. The same dense-solve scale caveat as PAC applies. This completes the PSS $\rightarrow$ PAC $\rightarrow$ Pnoise $\rightarrow$ PXF periodic small-signal suite.

⁵ PSP is ✓ since [Enhancement-132](#): a `.psp` command computes periodic small-signal S-parameters by exciting each RF port (`portnum/z0`, the `.sp` framework) through the same $(2M+1)N$ conversion matrix and forming $S^{\wedge}(k) = B^{\wedge}(k) \cdot A^{\wedge}-1$ per input frequency and conversion sideband $f_{\text{in}} + k \cdot f_0$. `pac_solve_at`'s matrix assembly is factored into a shared `pac_build_matrix`; the per-port solve drives each port's branch source ($V=1$) like `.sp`'s `VSRCspupdate`, and the power waves use the same Kurosawa convention, so $S = B \cdot A^{\wedge}-1$ is basis-invariant and reduces exactly to `.sp` for a time-invariant network. Verified 8/8: sideband-0 matches `.sp` to $\sim 10^{-16}$ for 1/2/3-port resistive and reactive networks (magnitude and phase) — including OSDI Verilog-A devices (conductance and reactive ddt stamps) — with correctly-zero conversion sidebands. Runs under both linear solvers (the conversion matrix is a standalone dense LU; PSS runs under both since E-118), verified bit-identical under KLU and Sparse.

⁶ QPSS is △(working two-tone, transient-sampling) since [Enhancement-133](#): a `qpss` command computes the two-tone steady-state spectrum — every mixing product $k_1 \cdot f_1 + k_2 \cdot f_2$ including IM3 — for commensurate tones (common beat $fb = \text{gcd}(f_1, f_2)$). It runs an ordinary transient over a few beat periods, then evaluates the Fourier coefficient directly at each exact intermod frequency (no FFT-bin rounding) and labels it by the 2-D index (k_1, k_2) . Solver-independent (drives a transient), works with built-in and OSDI devices. Verified 7/7 checks incl. the analytic IM3 products and the 3:1 IP3 slope law. [Enhancement-136](#) then added the frequency-domain two-tone Harmonic Balance engine (`qpss <expr> <f1> <f2> hb [K1] [K2]`): each node a 2-D Fourier series, devices sampled on a 2-D phase grid (θ_1, θ_2) (so incommensurate tones — irrational ratio, no common period — now work, which transient sampling cannot do) and 2-D DFT'd to the conversion matrix, Newton-solved by `pss_solve` with the E-135 source stepping; sources captured by an oversampled least-squares APFT. Verified 7/7 (analytic IM3 ratio, the incommensurate $\sqrt{2}$ case, $HB == \text{transient}$, $KLU == \text{Sparse}$) and it retains its operating point. [Enhancement-137](#) closes the gap (✓) with small-signal QPAC — the two-tone analogue of PAC: `qpac <f_in>` injects a small signal around the retained QPSS operating point and reports the response at every sideband $f_{\text{in}} + k_1 \cdot f_1 + k_2 \cdot f_2$, mixing it through the same 2-D conversion matrix (`qp_build_matrix` at f_{in} , solved by `pss_solve`) that the QPSS Newton used as its Jacobian. Verified 7/7, incl. reduce-to-AC (pump $\rightarrow 0$ the direct response equals the plain `.ac` and the sidebands vanish) and the v^2 -pump conversion ratio. [Enhancement-138](#) adds QPnoise (`qpnoise <output_node> <f_in>`), the two-tone analogue of `pnoise` on the same operating point: one ADJOINT solve $H^T \Psi = e_{\text{out}}, (0, 0)$ gives the transimpedance from every (node, sideband) to the output, and each device's DEVnoise folds $S \cdot |\Psi|^2$ over all sidebands (the mixer/PA noise conversion a static `.noise` cannot see). Verified 6/6, anchored by reduce-to-noise (pump $\rightarrow 0$ `onoise = plain .noise = 4kTR`, exactly). [Enhancement-139](#) adds the cyclostationary mode (`qpnoise ... cyclo`): the device PSD $S(t)$ swings over the two-tone period, so instead of a single-bias fold it averages over the 2-D phase grid, re-biasing each phase sample (with per-sample junction settling); it reduces to the stationary case by Parseval when S is constant. Both `qpnoise` modes were later corrected and hardened: E-177 fixed the folded-sideband source frequency $(|f_{\text{in}} + k_1 \cdot f_1 + k_2 \cdot f_2|)$, E-178 ported the exact separable cyclostationary folding to the 2-D grid $(B_{\{q1, q2\}})$ — verified digit-identical to the 1-D `pnoise cyclo` on the same circuit across the two orbit machineries — and its hardening pass replaced the old “ $\sim 8\times$ enhancement” hard-pumped-diode expectation, an artifact of uninitialized diode sidewall summary slots and the doubled HB DC bias (see note 7), with a closed-form torus-average referee the cyclo result now matches. [Enhancement-141](#) completes the two-tone small-signal suite with QPXF (`qpxf <output_node> <f_in>`), the ADJOINT of QPAC: one adjoint solve $H^T \Psi = e_{\text{out}}, (0, 0)$, each sideband block of Ψ dotted with the AC-source pattern, gives the transfer from an input at every sideband $f_{\text{in}} + k_1 \cdot f_1 + k_2 \cdot f_2$ to the output. By the reciprocity identity the sideband-(0,0) transfer is bit-identical to the QPAC response (verified 6/6). So

the quasi-periodic small-signal set now mirrors the single-tone PAC/Pnoise/PXF exactly: QPSS $\rightarrow$ QPAC $\rightarrow$ QPnoise $\rightarrow$ QPXF. [Enhancement-142](#) then gives all three a `dec|oct|lin` input-frequency sweep that emits a plottable ngspice plot (conversion gain / noise figure / image-rejection curves), matching how `.ac/.pnoise/.pxf` sweep — each swept point reuses the single-frequency solve and equals it to machine precision (5/5).

⁷ HB is  since [Enhancement-134](#): a `hb <f0> <K>` command solves the periodic steady state in the frequency domain by Newton -- each node voltage a truncated Fourier series, the KCL residual $F_k = I_{R,k}(V) + [dq/dt]_k - I_{s,k} = 0$ driven to zero with the E-121 $(2K+1)N$ conversion matrix as the exact Jacobian. The device residual/Jacobian are sampled by driving DC+AC loads at the current iterate's voltages; nonlinear reactive elements need NO charge extraction because $dq/dt = C(v)*v'$ -- the reactive current is the conversion matrix's $j\omega C$ term applied to V , using the sampled $C(t)$. Solver-independent (KLU + Sparse): the dense complex Newton is HB's own, so the linear solver is used only to read $G(t)/C(t)$ off the device matrix -- `hb_extract` carries the same `#ifdef KLU` complex-CSC binding as the PAC extraction, and `hb` honours `.option klu`, verified bit-identical under both. Built-in and OSDI devices. Verified 8/8 against the transient/fourier steady state, with quadratic Newton convergence: nonlinear R (analytic 3rd harmonic), nonlinear R+C, a real diode rectifier (junction devices are settled per sample so their limiter is a no-op), a compiled OSDI varactor whose $Q(v)$ 2nd harmonic matches, and $KLU==Sparse$ parity. Strongly-driven circuits (where a cold full-strength Newton diverges) are handled by automatic source-stepping continuation ([Enhancement-135](#)): every source is ramped by $\lambda: 0 \rightarrow 1$ in adaptive, warm-started, backtracking steps, so a 5 V diode rectifier that blows up cold converges in 3 continuation steps (easy circuits stay bit-identical). `.hb` and the rest of the HB family gained netlist dot-cards in [Enhancement-162/163](#). One serious latent defect was found and fixed by the [E-178](#) cross-machinery check: both HB Newtons (single-tone `hb` and two-tone `qpss ... hb`) double-subtracted the DC sources — the settle-mode rhs folded into the device-current term already carries them, and $-\lambda \cdot I_s$ subtracted them again — so every DC bias voltage converged to exactly $2\times$, silently corrupting all bias-dependent noise and conversion (flicker ∇I^{AF} off by 2^{AF}) while AC content and every AC-driven validation stayed exact. Fixed (net DC drive is now exactly $-\lambda \cdot I_{s_DC}$); `qpnoise cyclo` agrees with `pnoise cyclo` digit-for-digit across the two orbit machineries since. Multi-tone HB beyond two tones and a sparse block solve remain the follow-ups.

⁸ Envelope following is  since [Enhancement-154](#): the `envelope` command computes the slow amplitude/phase envelope of a carrier-driven circuit by sampling the state once per carrier period $T=1/f_c$ and integrating the slow drift, jumping M periods at a time. The per-period map is $X_{\{n+1\}} = \text{phi}(X_n)$ (phi = one carrier period of DAE integration); the naive forward-Euler envelope jump is unstable on high-Q circuits (the one-period monodromy has unit-circle eigenvalues) — which is exactly why an earlier forward-Euler attempt blew up and was shelved. E-154 uses the implicit backward-Euler jump $X_{\{n+M\}} = X_n + M(\text{phi}(X_{\{n+M\}}) - X_{\{n+M\}})$, Newton-solved with the finite-difference monodromy $\text{Phi} = d\text{phi}/dY$ as $[(1+M)I - M \cdot \text{Phi}] dY = -G$; the A-stable step tracks a resonator's envelope without diverging and its fixed point is the true steady state. Step size M is chosen by step-doubling LTE control; the one-period map reuses the transient primitives on a fixed `nppp` grid in trapezoidal mode (backward-Euler damps high Q), self-started. Verified against a full `.tran` under both solvers: $<3\%$ across a Q~3160 tank ring-up (26 envelope samples for ~3000 carrier periods), converging to the steady state, and ~1.6 % on a Q~316 tank. It pays off when the envelope is much slower than the carrier (high-Q / PLL / modulated-PA), where it is several times faster than the full transient; a second-order non-dissipative self-start and a sparse monodromy are follow-ups. With this, every analysis in the RF / periodic-steady-state suite is present.

Performance & scale

Category	Feature	ngspice	Spectre
Performance	Multithreaded device-model evaluation (OpenMP)		
Performance	Element bypass / latency exploitation		
Performance	Fast-SPICE hierarchical / isomorphic engine		
Performance	Distributed (MPI) / cloud partitioning		
Performance	Handles 10^6 – 10^8 -node post-layout netlists		

Statistical / variability / yield

Category	Feature	ngspice	Spectre
Statistical	Monte Carlo		
Statistical	Native process/mismatch modeling + correlations		
Statistical	Low-discrepancy sampling (Sobol / Latin-hypercube)		
Statistical	High-sigma methods (importance sampling, worst-case distance)		

Category	Feature	ngspice	Spectre
Statistical	Corner + MC + yield estimation flow	✓	✓

Monte Carlo was  (script-driven `alter + sgauss`); since [Enhancement-151](#) the `montecarlo` command packages the MC + spec + yield flow (Wilson-CI yield, optional LHS), so it is now ✓.

Low-discrepancy sampling is ✓ since [Enhancement-149](#): `mcsample lhs <N>` gives Latin-Hypercube stratification for the reset-driven `.param` Monte Carlo idiom (`agauss/gauss/aunif/unif/limit`), measured $\sim 130\times$ lower estimator variance at the same run count. Sobol and the nutmeg-loop `sgauss/sunif` idiom remain follow-ups.

High-sigma methods are ✓ since [Enhancement-150](#): `highsigma <N> -metric <expr> -max/-min <spec>` estimates 4–6 sigma rare-event failure probabilities by scaled-sigma importance sampling (inflate the Gaussian `.param` sigmas, reweight by the likelihood ratio) — direction-free, verified against the analytic $\Phi(-\beta)$ (e.g. a 5-sigma, $2.87e-7$ event recovered from ~ 6000 runs). Mean-shift / worst-case-distance importance sampling remains a follow-up.

Process/mismatch correlations and the corner+MC+yield flow are ✓ since [Enhancement-151](#): `mccorr` registers a correlation matrix and `mvnorm(i)` draws correlated process/mismatch factors (composing with LHS and importance sampling), and `montecarlo` packages the spec-based yield estimate (Wilson CI, optional LHS); corners are the ordinary `.lib` selection. A matched divider demo yields $\sim 100\%$ process-correlated vs $\sim 74\%$ independent — the correlation model is what decides it.

Reliability / aging

Category	Feature	ngspice	Spectre
Reliability	Device aging (HCI / NBTI / TDDDB)	✓ ¹⁰	✓
Reliability	Stress → degrade → re-simulate (fresh/aged) flow	✓ ¹⁰	✓
Reliability	Electromigration + IR-drop (EMIR)	✓ ¹¹	✓

Post-layout / parasitics

Category	Feature	ngspice	Spectre
Post-layout	Flat parasitic (RC) netlist simulation	✓	✓
Post-layout	RC reduction / model-order reduction	✓ ⁹	✓
Post-layout	n-port (S/Y/Z) extracted-block import	✓	✓

⁹ RC reduction is ✓ since [Enhancement-155](#): the `reduce` command collapses a post-layout parasitic R/C network into a small, electrically equivalent `.subckt` of R's and C's that preserves the port behaviour over DC. `.fmax`, using TICER (Time-Constant Equilibration Reduction) — Schur-complement elimination of interior nodes kept first-order in `s`, so the result is element-level R's and C's, not a model-order-reduction black box or passive-synthesis step (some TICER coupling caps can come out negative — electrically valid, not physically realizable). DC is preserved exactly; a factor knob trades reduction against in-band accuracy (monotone). Ports are auto-detected as nodes touched by any non-R/C device (sources, transistors, OSDI) plus ground and user keep nodes. Verified under both solvers: identity reduction is bit-exact, a moderate factor gives $\sim 4\times$ fewer nodes at <0.25 dB in-band error, and an OSDI device auto-marks its port. [Enhancement-156](#) then makes the engine sparse — per-node adjacency lists + minimum-degree elimination (like sparse LU) + a `maxdeg` fill guard — lifting the node cap from ~ 2500 into the millions (a 65k-node network reduces in ~ 4 s), so it reaches real extraction scale.

¹⁰ Device aging and the stress → degrade → re-simulate flow are ✓ since [Enhancement-157](#): the `aging <t_target>` command finds every aging-capable device, computes how much it has degraded after the target lifetime, writes that back, and re-stamps the circuit so any later analysis sees the aged devices (fresh vs aged). It is model-agnostic — a Verilog-A/OSDI model opts in by exposing a degradation-rate operating-point variable (`agerate`) and a per-instance `age` parameter; the engine integrates the rate into a dose and feeds it back, the model owns the physics (dose → parameter shift). Static mode uses the DC operating-point stress; dynamic mode time-averages the rate over a transient (capturing duty cycle). Verified under both solvers against an NBTI demo NMOS: exact dose, the analytic $\Delta V_{th} \propto t^{\theta} \cdot 0.25$ power law, monotone degradation, near-threshold sensitivity, and $0.30\times$ aging for a 30%-duty gate.

¹¹ Electromigration + IR-drop (EMIR) is ✓ since [Enhancement-158](#): the `emir` command runs a DC solve of the power-distribution network and reports IR-drop (how far each node sags below the supply rail under load, worst node + violations past a threshold) and electromigration (per wire-segment current density $J = |I|/(w \cdot \text{thickness})$, ranked, with a Black's-equation relative lifetime $MTTF/ref = (J_{max}/J)^n$). The point of the analysis is that EM is set by current density, not current — a fat trunk carrying huge current is safe while a thin wire at a fraction of that current voids first — which is why real grids taper wire width with carried current. Verified under both solvers on a tapered ladder: exact IR-drop, linear scaling, $J=I/(w \cdot \text{thick})$ exact, narrow-wire-is-worst physics, Black J^{-n} scaling, and an OSDI current load. First cut is DC/average-current; transient/RMS EM and a temperature-coupled MTTF map are follow-ups.

Behavioral / mixed-signal / AMS

Category	Feature	ngspice	Spectre
Modeling	Verilog-A (analog, LRM Annex C) via OSDI	✓	✓
Modeling	XSPIICE code models + event-driven digital	✓	✓
Modeling	Verilog-AMS mixed-signal (connect modules)	✗	✓
Modeling	Real-number modeling (wreal / RNM)	✗	✓
Modeling	VHDL-AMS	✗	✓

Interfaces & infrastructure

Category	Feature	ngspice	Spectre
Interface	Shared-library / programmatic API	✓	✓
Interface	Python bindings	✓	✓
Infrastructure	Built-in optimizer	✓	✓
Infrastructure	Checkpoint / restart of long runs	✓	✓

The “built-in optimizer” row is ✓ since [Enhancement-130](#): the `optimize` command is a derivative-free Nelder-Mead search that varies device/`alter` parameters, re-runs an analysis, and minimizes an objective expression in normalized $[0,1]$ space — verified to reach analytic optima in 1-D and 2-D. [Enhancement-143](#) adds a gradient-based least-squares mode: one or more weighted `-target <expr> <value>` measurements, optionally spread over several `-analysis` stages (so a single fit can combine, say, a DC operating point and an AC response), are fitted with Levenberg-Marquardt (finite-difference Jacobian) — the right tool for smooth curve-fitting / device-parameter-extraction problems, reaching the optimum in far fewer analysis runs than the simplex (e.g. 27 vs 67 evaluations on a two-target RC fit). Verified to recover both i s and n of a compiled OSDI diode from two I-V points. [Enhancement-144](#) closes the last gap in the knob set: besides `alter-reachable` device/instance parameters (`-param`), the optimizer can now tune symbolic netlist `.param` values (`-dparam`) — since those are expanded at parse time, each candidate is applied with `alterparam` and a quiet reset that re-sources the deck (re-evaluating the `.param` expressions and re-stamping device values). Deck params are re-sourced first and the in-place `alter` params re-applied after, so the two kinds mix in one run; verified on `.params` used directly and inside expressions, including an OSDI-device value. [Enhancement-145](#) adds the third knob kind, `-mparam`, for `.model-card` parameters (named `@<model>[<param>]`): a model parameter is not `alter-reachable` (only sources, resistors and device instance parameters are — which is also exactly what `.dc` can sweep), so it is changed in place with `altermod` (no re-source, unlike `.param`). All three kinds — `-param` (instance/`alter`), `-mparam` (model/`altermod`), `-dparam` (`.param`/re-source) — mix in one optimization; verified recovering an OSDI and a built-in diode model parameter and a joint model+instance fit. [Enhancement-146](#) reuses the same knob-application machinery for a universal parametric sweep — a sweep command and `.sweep` card that step any knob (auto-detecting instance / model / `.param`) over a range, run a chosen inner analysis at each point and record the outputs into a plot. This is the parametric-analysis / `.step` capability commercial tools have: `.dc` can only step sources, resistors and instance parameters, whereas sweep also covers model parameters and symbolic `.params`. Verified to reproduce the built-in `.dc` bit-for-bit on an instance sweep and to match the analytic response on model-parameter, `.param`, AC and transient sweeps.

The “checkpoint / restart” row is ✓ since [Enhancement-131](#): stock ngspice could only continue a paused run in memory (`stop/resume`); the new `savestate <file> / loadstate <file>` commands serialize the full transient integration state (solution vector, device state history, time/step/order, pending breakpoints) to disk and resume it — including in a fresh process — so a long run survives a crash, splits across sessions, or moves between machines. The resumed waveform is bit-identical to an uninterrupted run for built-in devices. Since [Enhancement-180](#) it works under both linear solvers — and across them (save under one, resume under the other): the E-131 KLU guard had masked a solver-mode ordering bug in `loadstate`, not a real limitation.

Where to invest (given the Verilog-A/OSDI side is done)

The realistic, winnable identity for this project is the open-source, OSDI/Verilog-A-native simulator with a real RF-noise and statistical story — not out-engineering 25 years of commercial fast-SPICE/parallel work. Ranked by leverage on the existing strength $\times$ differentiation $\times$ tractability:

1. RF periodic small-signal suite — PAC $\rightarrow$ Pnoise $\rightarrow$ PXF, on the hardened PSS. The PSS foundation is now shipped and verified under both linear solvers ([Enhancement-117](#), [Enhancement-118](#)), and [Enhancement-119](#) retains the periodic operating point (voltages + device states per sample) and [Enhancement-120](#) turns it into the periodic Jacobian harmonics G_k , C_k , and [Enhancement-121](#) assembles those into the $(2M+1)N$ harmonic conversion matrix and solves it, [Enhancement-122](#) wraps that in a working `.pac` command, and [Enhancement-123](#) finishes it with a netlist-referenced source stimulus and multi-sideband conversion-gain output — a complete periodic-AC analysis, verified against the exact linear transfer and driving-point responses, [Enhancement-124](#) adds `.pnoise` — folding every device's noise through the conversion-matrix adjoint H^T (reusing the existing device noise routines), verified to reduce exactly to `.noise` in the linear limit — and [Enhancement-125](#) adds `.pxf`, the adjoint transfer function, whose sideband-0 result is bit-identical to the PAC response by reciprocity. The PSS $\rightarrow$ PAC $\rightarrow$ Pnoise $\rightarrow$ PXF periodic small-signal suite is now complete — genuinely novel in open source — and [Enhancement-126](#) adds cyclostationary noise (per-sample bias, time-domain-transfer fold) so pumped devices' bias-dependent noise is handled correctly. The single-tone suite is now complemented by frequency-domain Harmonic Balance ([Enhancement-134](#)), oscillator phase noise ([Enhancement-140](#)), the two-tone quasi-periodic set (QPSS/QPAC/QPnoise/QPXF, [Enhancement-136–142](#)), and finally envelope following ([Enhancement-154](#), implicit monodromy period-jumping) — so every analysis in the RF / periodic-steady-state suite is now present. The suite has since been through a systematic correctness-audit arc ([E-171–178](#): KLU pole-zero determinants, the conversion-matrix parametric term, driven-mode PSS, the noise-folding referee, exact cyclostationary folding, and the HB DC-source fix — each found by probing a region no prior test could see). What remains is efficiency refinement (sparse conversion-matrix and monodromy solves, three-plus-tone) rather than new analyses.
2. Convergence robustness — coordinated accuracy presets (`errpreset`) landed in [Enhancement-110](#), a globalized Newton line search in [Enhancement-111/112](#), and pseudo-transient continuation (`.option ptcont`) in [Enhancement-127](#) — so the principled globalization and continuation gaps are now closed. What remains is mostly auto-triggering heuristics (reaching for these aids without the user asking) and folding them into the robustness presets. Self-contained in the ngspice core.
3. Statistical sampling — delivered. The whole statistical column is now ✓: Latin-Hypercube low-discrepancy sampling ([Enhancement-149](#), `mcsample lhs`, $\sim 130\times$ lower estimator variance), the high-sigma rare-event tail ([Enhancement-150](#), `highsigma` scaled-sigma importance sampling, 4–6 sigma from a few thousand runs), and native process/mismatch correlations plus a packaged yield flow ([Enhancement-151](#), `mccorr/mvnorm` • `montecarlo`). What remains is only efficiency variants (mean-shift / worst-case-distance importance sampling) and a one-command corner-sweep-of-yield wrapper.

Explicitly lower priority: fast-SPICE parallelism and aging/EMIR — enormous efforts that don't leverage what makes this project distinctive.